

GATE-AII BRANCHES Engineering Mathematics



Linear Algebra



Lecture No.- 6

By- Rahul sir

Topics covered in previous lecture



Topic

Properties of matrices

Topic

Question based on properties of the matrix

Topics to be Covered



Topic

Properties of matrices

Topic

**Question based on properties of the matrix,
Orthogonal basis**

Topic

**Linear dependence and independence,
Rank space and null space**

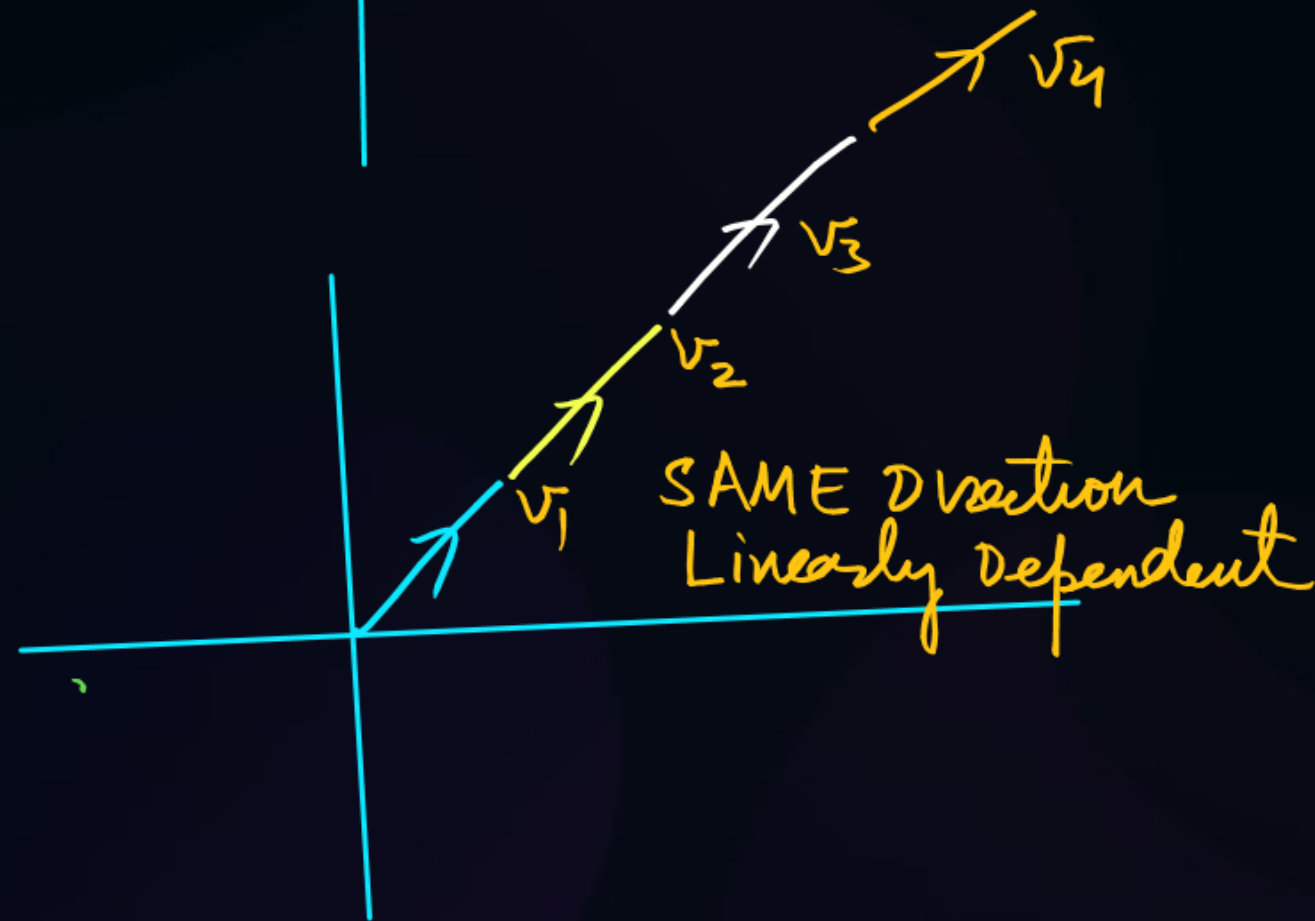
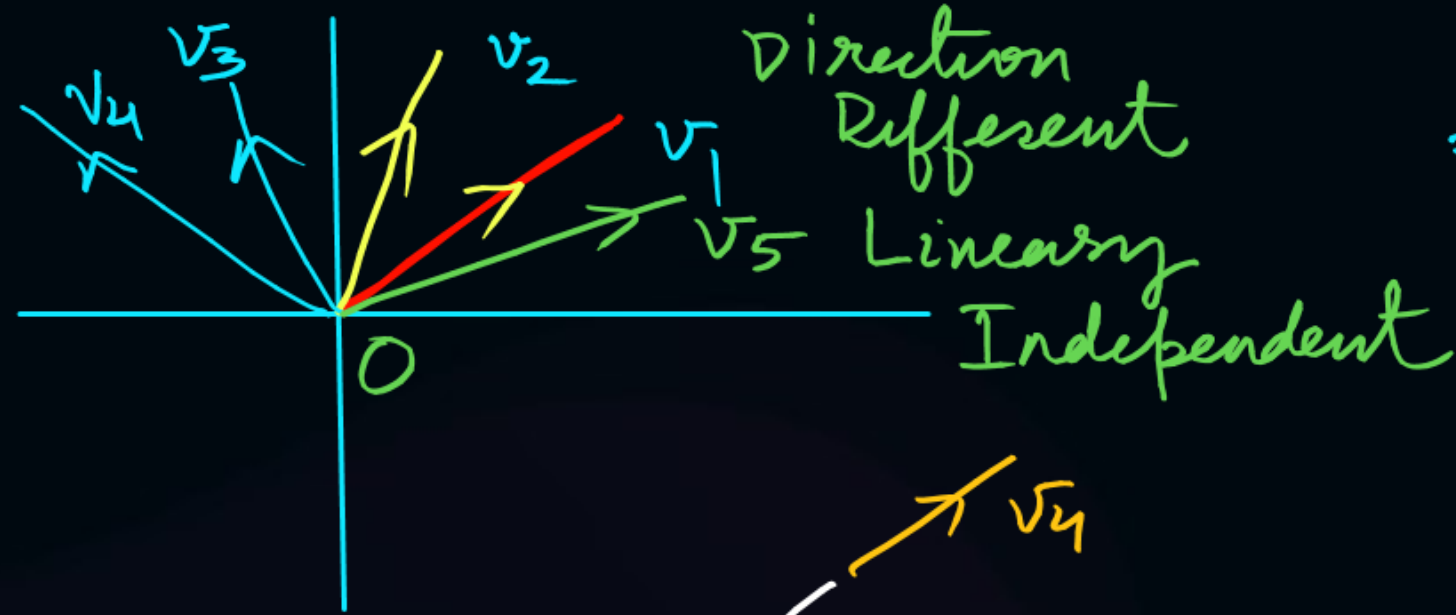
Topic

System of equations

Topic

**Question based on system of equations,
Span of vectors**

Rank of Matrix :



No of Linearly Independent Vectors
OR
(Row/column) ✓

No of Non-ZERO rows
(Echelon Form)
Any matrix $m \times n$] Non square / square matrix

System of Equations

$$\begin{cases} a_{11}x + a_{12}y + a_{13}z = b_1 \text{ or } 0 \\ a_{21}x + a_{22}y + a_{23}z = b_2 \\ a_{31}x + a_{32}y + a_{33}z = b_3 \end{cases}$$

3d space - 3d Planes

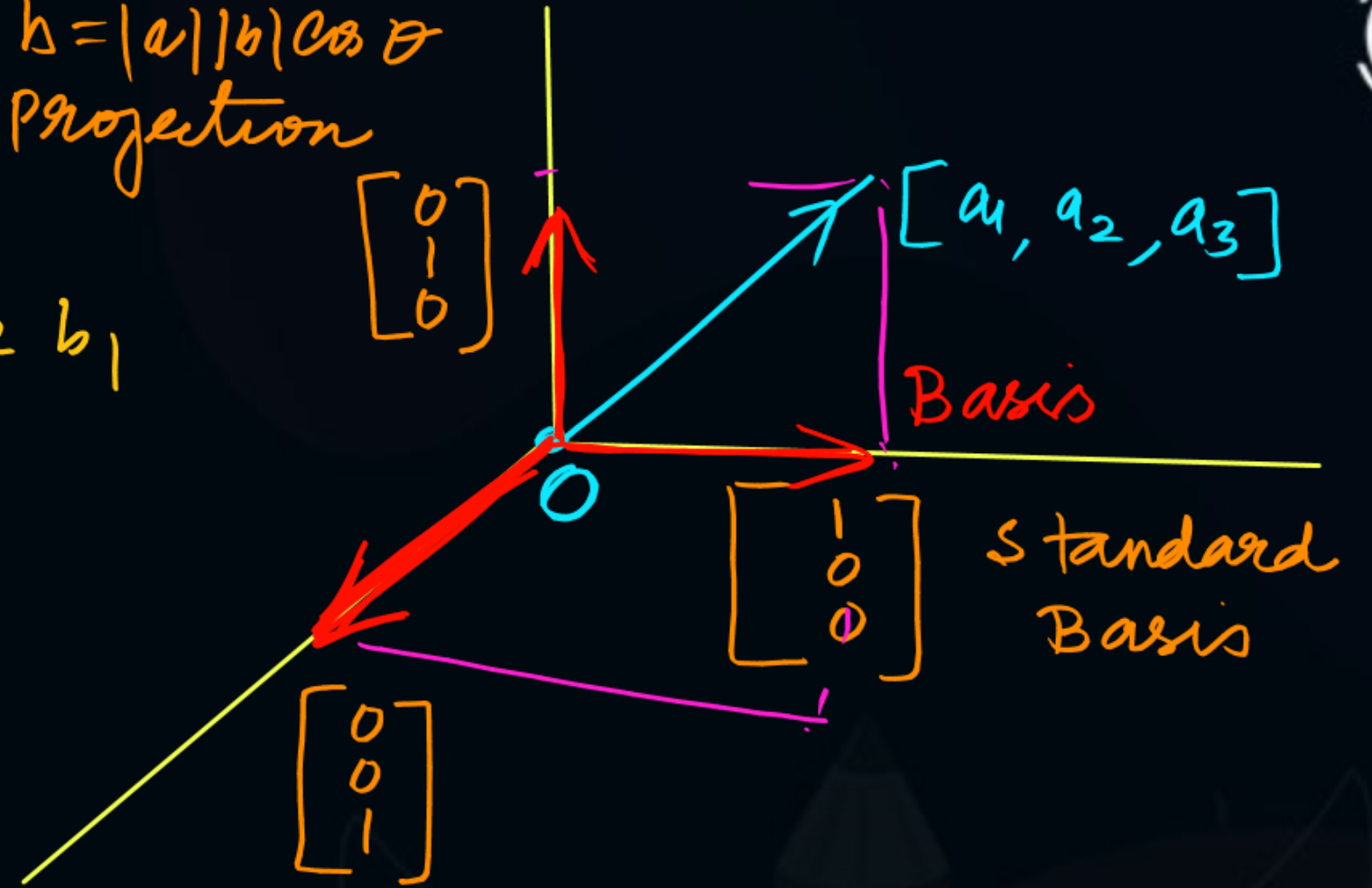
$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \end{bmatrix}$$

$$\text{Dot Product} = 0 \neq 0$$

$$a \cdot b = |a||b|\cos\theta$$

Projection



$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3}$$

Square matrix

Non Square matrix

$m = \text{Rows}$ $m \times n$
 $n = \text{columns}$

= Elimination Method.
 = No. of Non-Zero Rows

$x + y = 5$
 $2x + 2y = 10$

$$\det A = \begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} = 0$$

$$\det A = 0 \leftarrow \text{No. of}$$

Linearly Independent vectors

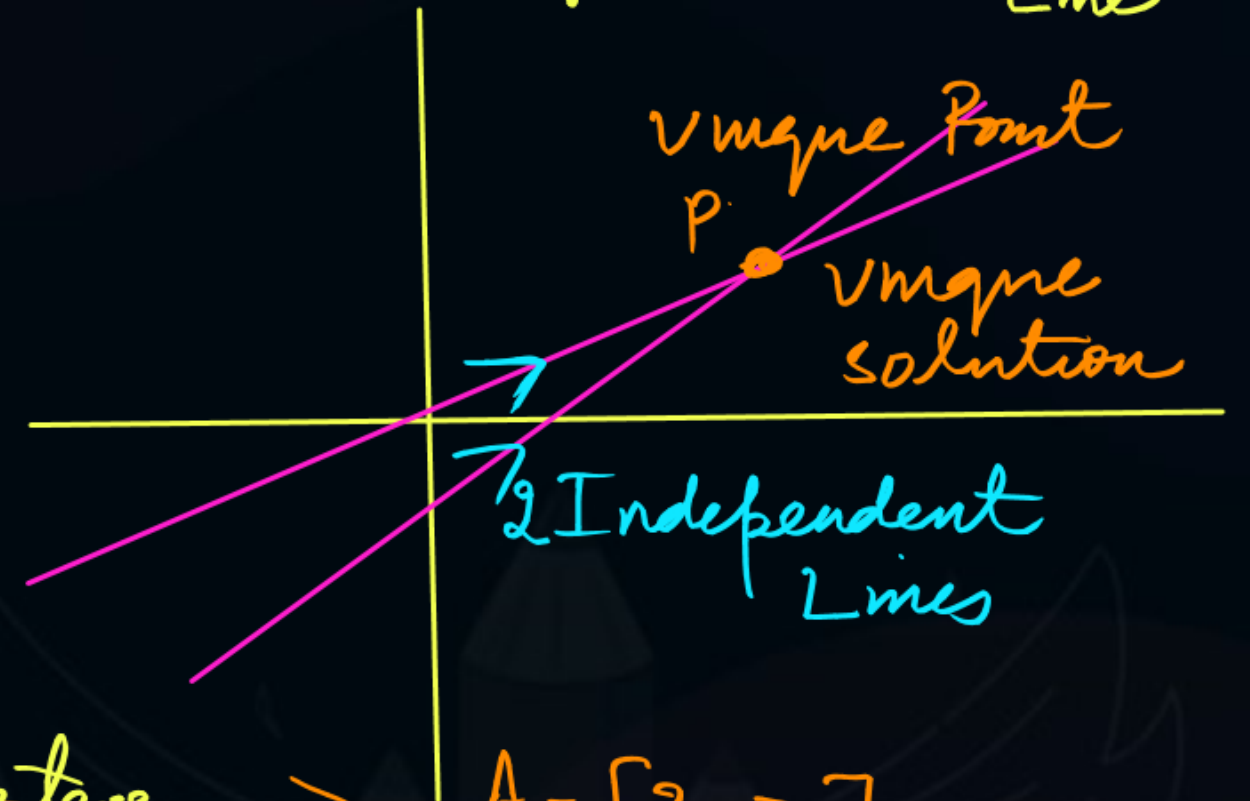
$\det A \rightarrow$ No. of linearly Independent vectors



Parallel Lines

Infinite solⁿ

$2x + 3y = 5$
 $x + 13y = 10$] straight Line



Unique Point P

Unique solution

2 Independent Lines

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 13 \end{bmatrix} \det A \neq 0$$

$\det A \rightarrow$ No. of L.I.d $\rightarrow$ rank

Any Square matrix

If any $n \times n$ matrix

For 3×3 matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3}$$



(A) $\det A \neq 0, |A| \neq 0$

$$r(A) = \rho(A) = 3$$

$$\boxed{r(A) = \rho(A) = n}$$

Rank of matrix = No of variables = 3

1) Null matrix $\boxed{f(A) = 0}$

(B) $\det A = 0, |A| = 0$

Rank of matrix < no of variables.

Decrease The
No. of Independent Planes.



$|A| = 0$ $\rho(A) < \text{no of variables}$

$\rho(A) = 2$ If all subminors Are Not ZERO.

$|A| = 0$ $\rho(A) < \text{no of variables}$

"All subminors Are ZERO" No. of Independent Planes / vectors / Row $< n$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$$

$$\boxed{\rho(A) = 1}$$

$$0 = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix} = 0$$

$$0 = \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix} = 0$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & 5 \\ 2 & 15 & 16 \end{bmatrix}_{3 \times 3} \quad |A| \neq 0 \quad (3)$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 2 & 15 & 16 \end{bmatrix} \quad \frac{\det A = 0}{\rho(A) = 2}$$

$$0 = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix} = 0$$

$$\neq 0 \begin{bmatrix} 1 & 2 \\ 2 & 15 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 15 & 16 \end{bmatrix} \neq 0$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 16 \end{bmatrix} \neq 0$$

Non-Square matrices

$$A \rightarrow 3 \times 4$$

$$A = \begin{bmatrix} c_1 & c_2 & c_3 & c_4 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 7 & 9 & 10 \end{bmatrix} \begin{matrix} - R_1 \\ - R_2 \\ - R_3 \end{matrix}$$

3×4

Using elimination

$$R_2 \rightarrow R_2 - 2R_1$$

→ my strategy is clear → Row echelon form
 $\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 1 & 7 & 9 & 10 \end{bmatrix} 3 \times 4$$

$$\boxed{\rho(A) = 2}$$

→ No of Non-ZERO rows.]



Topic: Vector space

$(n+1) \times (n+1)$
matrix

#Q. The rank of the following $(n+1) \times (n+1)$ matrix, where 'a' is a real number

is 1

$$\begin{matrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 - R_1 \\ \vdots \\ R_n \rightarrow R_n - R_1 \end{matrix} \begin{bmatrix} 1 & a & a^2 & \dots & a^n \\ 1 & a & a^2 & \dots & a^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a & a^2 & \dots & a^n \end{bmatrix} \begin{matrix} -R_1 \\ -R_2 \\ -R_3 \\ -R_4 \\ \vdots \\ -R_n \end{matrix}$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$

$$= \rho(A) = 1$$

A

1

B

2

C

n

D

Depends on the value of a

$$\begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & 1 & 1 & \dots & 1 \\ 1 & 1 & 1 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & 1 \end{bmatrix} \begin{matrix} -R_1 \\ -R_2 \\ -R_3 \\ \vdots \\ -R_n \end{matrix}$$



Topic: Vector space

#Q. The rank of the matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

is

for 3×3
✓ Calculate $\det A = 0$
✓ Check subminors.
 $\boxed{\rho(A) = 2}$

A

0

B

1

C

2

D

3



Topic: Vector space

Physical Signi



#Q. Let $A = [a_{ij}]$, $1 \leq i, j \leq n$ with $n \geq 3$ and $a_{ij} = i \cdot j$, Then the rank of A is

$$\boxed{\rho = 1}$$

$i \cdot j$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\boxed{\text{Det } A = 0}$$

$$\boxed{\rho(A) = 1}$$

A 0

B 1

C $n-1$

D n

$$a_{ij} = i \cdot j$$

$$a_{32} = 3 \times 2$$

$$a_{33} = 3 \times 3$$

$$a_{21} = 2 \times 1$$

$$a_{22} = 2 \times 2$$

$$a_{23} = 2 \times 3$$

$$a_{31} = 3 \times 1$$

$$a_{11} = 1 \times 1$$

$$a_{12} = 1 \times 2$$

$$a_{13} = 1 \times 3$$



Topic: Vector space

$$[x_1, x_2, x_3, \dots, x_n]$$

#Q. $X = [x_1 \ x_2 \ \dots \ x_n]^T$ is an n -tuple non-zero vector. The $n \times n$ matrix $V = XX^T$

n Tuple $[x_1, x_2, \dots, x_n]$
 $X = [x_1, x_2, x_3]$ $n=3$ Tuple

$$X = [x_1 \ x_2 \ x_3]$$

$$X^T = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$V = [XX^T]$$

$$= \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

1×3 3×1

A

Has rank zero

B

Has rank 1

C

Is orthogonal

D

Has rank n

$\text{rank} = 1$



Topic: Vector space



#Q. $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}; B = \begin{bmatrix} p^2 + q^2 & pr + qs \\ pr + qs & r^2 + s^2 \end{bmatrix}$

rank of matrix B is

A $N/2$

B $N - 1$

C N ✓

D $2N$

$$= p^2 s^2 + q^2 r^2 - 2 p s r q - (ps - qr)^2 = \textcircled{N} \checkmark$$

If the rank of matrix A is N, then the

$$A = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \Rightarrow \begin{bmatrix} ps - qr & r \end{bmatrix} = \textcircled{N}$$

$$B = \begin{bmatrix} p^2 + q^2 & pr + qs \\ pr + qs & r^2 + s^2 \end{bmatrix}$$

→ Det B

$$= (p^2 + q^2)(r^2 + s^2) - (pr + qs)^2$$

$$= \cancel{p^2 r^2} + \cancel{p^2 s^2} + \cancel{q^2 r^2} + \cancel{q^2 s^2}$$

$$- [\cancel{p^2 r^2} + \cancel{q^2 s^2} + 2 p q r s]$$



Topic: Vector space



#Q. The rank of the matrix

$$\begin{pmatrix} 6 & 0 & 4 & 4 \\ -2 & 14 & 8 & 18 \\ 14 & -14 & 0 & -10 \end{pmatrix} \text{ is}$$

$$\begin{array}{l} \rho(A) = 2 \\ \hline \text{CE} \end{array}$$

System of Equations:

$$a_{11}x + a_{12}y + a_{13}z = 0$$

$$a_{21}x + a_{22}y + a_{23}z = 0$$

$$a_{31}x + a_{32}y + a_{33}z = 0$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\boxed{AX = B}$$

System of Equation

$AX = 0$ (Homogeneous eqnⁿ)

= Natural Oscillations

= Transient state

= No Force apply

= No Battery apply

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

= Non Homogeneous eqnⁿ

= forced oscillations

= Steady state

$$AX = B$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

3 planes

Augmented matrix

$$C = [A : B]$$

$$C = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right]$$

— R_1

— R_2

— R_3

using
elimination

$$C (3 \times 4)$$

$$\begin{cases} x + y + z = 5 \\ x + 6y + 9z = 45 \\ 13x + 91y + 101z = 102 \end{cases}$$

3 Independent
Planes

$$\rho(A) = 3$$

$$C = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 1 & 6 & 9 & 45 \\ 13 & 91 & 101 & 102 \end{array} \right]$$

$$\rho(A) = 3$$

no of variables = 3

Unique solution

$$\rho(A) = \rho(C) = \text{number of variables}$$

Single
Unique
solution

$x + y + z = 5$
 $2x + 2y + 2z = 10$
 $x + 5y + 9z = 10$
 $\rho(A) = 2$

$$C = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 2 & 2 & 2 & 10 \\ 1 & 5 & 9 & 10 \end{array} \right] \quad \rho(C) = 2$$

2 Independent Planes

Infinite solution

$$\rho(A) = \rho(C) < \text{no of variables}$$

Infinite
soln

No solution

Rank $A \neq \text{rank}(C)$ (Parallel Planes)

$$\rightarrow \text{rank} = 2 \quad \text{rank } C = 3$$

$$\underline{2 \neq 3}$$

System of Equⁿ $AX=B$

Consistent

Unique solⁿ
 $\rho(A) = \rho(C) = n$

Infinite
 $\rho(A) = \rho(C) < n$

No solⁿ
 $\rho(A) \neq \rho(C)$

Always gives a solution

Inconsistent
 No solution

THANK - YOU